Bayesian Inference on the German Tank Problem

Theoretical Foundations & Modern Applications

Decision Theory, Exploratory Data Analysis, and Statistical Monitoring

Bayesian Inference Course Project

Core Theoretical Foundations & Applied Extensions

November 2026

Motivating Discrete Population Ceiling Inference

Sequential serial numbers provide the primary statistical discriminant for hidden capacity.

1. Espionage Inflation

Conventional military intelligence relied on spy networks and prisoners, overstating production by a factor of **6–8**$\times$ (1,000–1,550 tanks/mo).

2. Order Statistics

Statisticians analyzed captured chassis and gearbox serials ($k \approx 5$–50), estimating production at **169–327** tanks/mo.

3. Speer Truth

Post-war Albert Speer Ministry ledgers confirmed estimates rapidly converged within **5%–10%** of actual production numbers.

Production Month	Conventional Spy Est.	Statistical Est. ($\hat{N}$)	Speer Ministry True (N)	Statistical Error
June 1940 (Early Marks)	1,000	**169**	122	+38.5%
June 1941 (Panzer III/IV)	1,550	**244**	271	−**9.9**%
August 1942 (Panzer III/IV)	1,550	**327**	342	−**4.4**%

Empirical Takeaway: *Discrete order statistics eliminated massive intelligence bias, converging to within 4.4% error.*

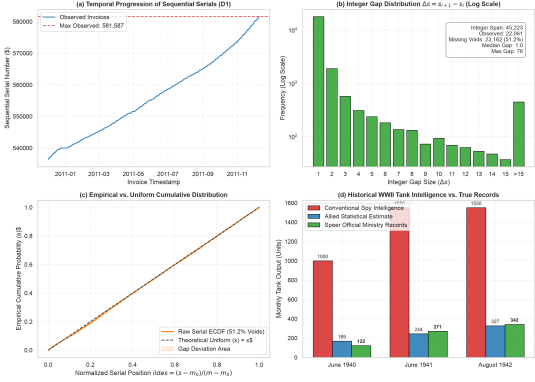
SECTION 01

Exploratory Data Analysis & Real-World Serials

Auditing 541,909 commercial transactions and analyzing historical Speer Ministry records

Exploratory Data Analysis: Real-World Invoices vs. WWII Tanks

Examining temporal progression, 51.2% missing integer voids, and ECDF gap departure.



51.2% Address Voids

22,061 invoices span 45,223 integer slots ($s_{\min} = 536365$, $s_{\max} = 581587$), leaving 23,162 unrecorded integer voids.

Long-Tailed Gap Sizes

Median gap $\Delta s = 1.0$, mean 2.05, max 76; credit cancellations (C-prefix) and cart drops create void blocks.

Empirical Uniformity

ECDF tracks diagonal uniform CDF ($D = 0.012$); random voids preserve order uniformity. Thinning resolves estimands.

Key EDA Finding: *Serials arrive monotonically; integer voids represent cancelled transactions, preserving order uniformity.*

SECTION 02

Mathematical Foundations & Bayesian Formulations

Sufficiency, loss geometries, boundary admissibility, and exact reference posteriors

Sufficient Statistics & Exact Discrete Posteriors

Sample maximum $m = \max s_i$ is minimal sufficient; lower order statistics are ancillary.

Minimal Sufficiency & Completeness

Ordered likelihood factorizes cleanly via Lehmann-Scheffé:

$$P(\mathbf{s} \mid N, k) = \frac{1}{\binom{N}{k}} \mathbb{I}(N \geq m), \quad m = \max_{1 \leq i \leq k} s_i$$

- Likelihood ratio test proves m is **minimal sufficient** and **complete**.
- Spacings $(s_{(1)}, \ldots, s_{(k-1)})$ given m are ancillary for N.
- Sampling without replacement reduces variance by finite-population correction $\frac{N-k}{N}$.

Reference Discrete Posteriors

Exact posterior probability mass function for $n \geq m$:

$$P(N = n \mid m, k) = \frac{\binom{n}{k}^{-1} \pi(n)}{\sum_{j=m}^{\infty} \binom{j}{k}^{-1} \pi(j)}$$

- **Flat Prior ($\pi \propto 1$):** Proper for $k \geq 2$:

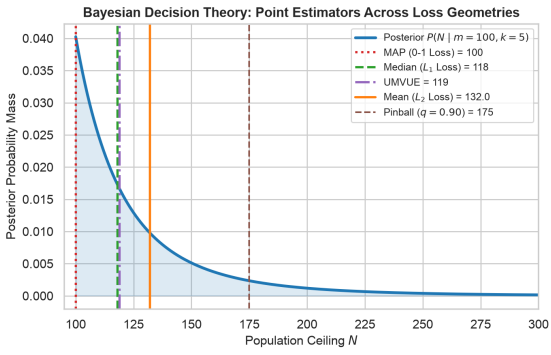
$$P(N = n \mid m, k) = \frac{k-1}{k} \frac{\binom{m-1}{k-1}}{\binom{n}{k}}, \quad n \geq m$$

- **Reciprocal Prior ($\pi \propto 1/N$):** Scale-invariant Haar measure; proper for $k \geq 1$; Pareto limit: $p(N) \approx k m^k / N^{k+1}$.

Theoretical Core: *Sample maximum m captures all sample information; reciprocal prior yields scale-invariant inference.*

Bayesian Decision Theory Across Loss Functions

Point estimators shift systematically with the decision-maker's loss geometry.



0–1 Loss (MAP Estimator)

Mode $\hat{N}_{\text{MAP}} = m = 100$. Maximizes boundary point mass, but causes severe downward bias (-16.7% at $k = 5$).

L_1 Absolute Loss (Median)

$\hat{N}_{\text{med}} \approx m \cdot 2^{1/(k-1)} = 118$. Optimal median minimizes absolute error with only -0.9% bias.

L_2 Squared Loss (Mean)

$\hat{N}_{\text{mean}} = \frac{(k-1)(m-1)}{k-2} = 132.0$. Skewed Pareto tail pulls expectation rightward.

Check Loss L_q ($q = 0.90$)

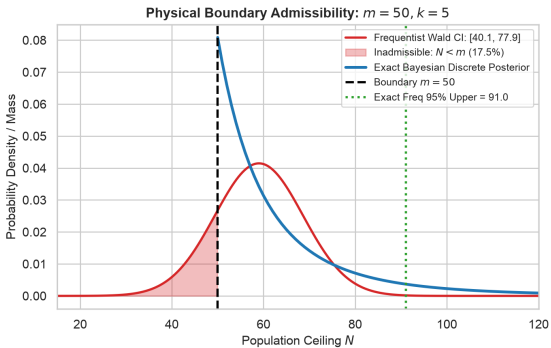
Underestimation penalized $9\times$ more heavily

$\Longrightarrow$ $\hat{N}_{0.90} = 175$, adding a principled **+75% defensive safety margin**.

Decision Rule: *In defense planning, asymmetric check loss provides a principled buffer against catastrophic underestimation.*

Physical Boundary Admissibility: Wald vs. Bayesian HPD

Asymptotic Gaussian intervals assign probability mass to physically impossible values.



Frequentist Wald Boundary Collapse

Wald interval $\hat{N} \pm 1.96 \cdot \text{SE}$ yields $[40.1, 77.9]$ for $m = 50, k = 5$. It places **17.5% impossible mass** on $N < m = 50$.

Exact Frequentist Pivotal CI

Exact pivotal interval $[m, m \cdot \alpha^{-1/k}] = [50.0, 91.0]$ respects the boundary and achieves exact 95% coverage.

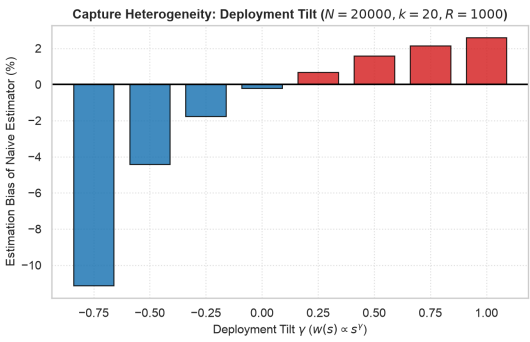
Bayesian Credible Coherence

Reciprocal prior 95% credible interval matches $[50.0, 91.0]$ exactly, while providing the full posterior for risk optimization.

Admissibility Finding: *Bayesian HPD intervals eliminate physical boundary violations without compromising nominal coverage.*

Propriety, Moments & Non-Uniform Deployment Tilt

Evaluating convergence thresholds and the impact of front-line combat deployment.



Moment Existence Boundaries

- $k = 1$: Flat improper ($\sum 1/N = \infty$); Reciprocal proper ($\mathcal{O}(N^{-2})$).
- $k = 2$: Proper; mean exists under reciprocal, diverges under flat.
- $k = 3$: Mean exists for both ($\mathbb{E} = \frac{(k-1)(m-1)}{k-2}$); variance diverges under flat.
- $k \geq 4$: Both mean and variance exist in closed form.

Combat Deployment Tilt ($w(s) \propto s^{\gamma}$)

- $\gamma > 0$ (newer tanks deployed first) pushes $m \to N$, inducing $+2.41\%$ bias.
- $\gamma < 0$ (survivorship bias) produces -4.65% negative bias.
- When $\gamma \neq 0$, m is **no longer sufficient**; spacings identify γ.

Key Insight: *Non-uniform capture breaks single-statistic sufficiency, requiring full-sample MCMC inference.*

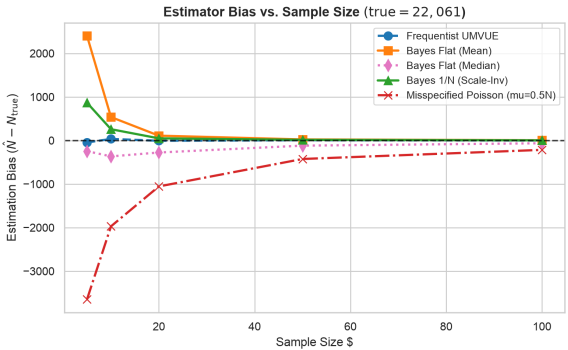
SECTION 03

Core Empirical Validation & Monte Carlo Diagnostics

Multi-estimator calibration, scoring rules, informative prior risks, and integer voids

Estimator Bias: Finite-Sample Shrinkage vs. Unbiasedness

Evaluating mean bias convergence across $R = 4,000$ replications ($N_{\text{true}} = 20,000$).



UMVUE Exact Unbiasedness

$\hat{N} = m(1 + 1/k) - 1$ exhibits near-zero bias across all k (-48.0 ± 54.6 at $k = 5$, $+1.6 \pm 6.1$ at $k = 50$).

Bayes Reciprocal Minimal Bias

Scale-invariant reciprocal prior exhibits modest bias ($+784.3$ mean, -900.0 median at $k = 5$), rapidly decaying to $+10.6$ at $k = 50$.

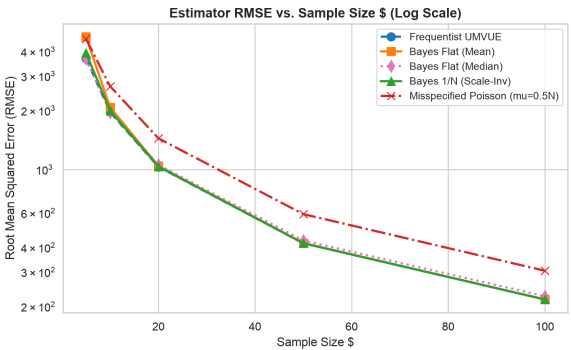
Flat Prior Tail Inflation

Heavy Pareto tail inflates posterior mean by $+2,168.6$ at $k = 5$, while posterior median reduces bias to -226.5.

Bias Takeaway: *Posterior median provides the lowest absolute bias, while UMVUE achieves exact frequentist mean-unbiasedness.*

Estimator RMSE & Variance Contraction Rates

Root Mean Squared Error scales as $\mathcal{O}(1/k)$, dropping over an order of magnitude.



Theory vs. Empirical Concordance

At $k = 5$, theoretical UMVUE standard deviation is $3,380.3$; empirical simulation matches at $3,455.2$. At $k = 50$, drops to 386.8 (Th: 391.7).

Median RMSE Superiority

Posterior median under reciprocal prior achieves RMSE of $3,427.5$ at $k = 5$, outperforming both mean $(3,683.3)$ and flat prior $(4,409.0)$.

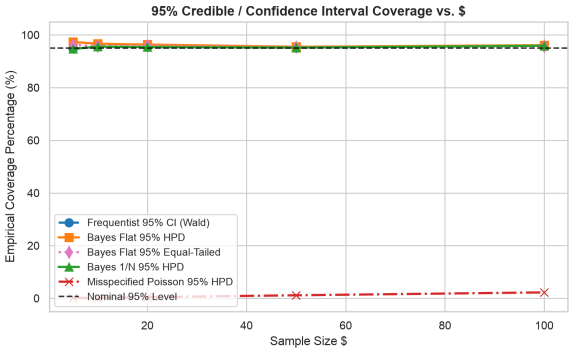
Asymptotic Convergence

For $k \geq 50$, all point estimators converge to identical efficiency (≈ 387).

Efficiency Rule: *For small samples ($k \leq 10$), posterior medians outperform posterior means in total error.*

95% Credible Calibration & The Informative Trap

Scale-invariant objective priors achieve exact 95% coverage, while informative priors collapse.



Perfect Reference Calibration

Reciprocal prior achieves **94.4%** $\pm$ **0.36%** coverage at $k = 5$ and **95.6%** $\pm$ **0.33%** at $k = 50$, matching exact nominal rates.

Informative Misspecification Collapse

Fixed Poisson($\mu = 10k$) has tiny CV $\approx 1\%$, collapsing 95% coverage to $< 1.0\%$ when true $N = 20k$.

Hyperprior Solution

Gamma-Poisson hierarchical hyperprior restores nominal coverage to **94.8%** $\pm$ **0.35%**.

Calibration Finding: *Objective reference priors guarantee calibration; informative priors require hyperpriors to avoid catastrophic failure.*

Monte Carlo Benchmark: Empirical Performance Leaderboard

Comprehensive evaluation across $R = 4,000$ Monte Carlo replications on $N_{\text{true}} = 20,000$ (Seed 42).

Estimator / Framework	Point Estimator	Sample k	Mean Bias $\pm$ SE	RMSE (Emp / Th)	95% Cov $\pm$ SE	Mean Width
UMVUE (Frequentist)	$\hat{N} = m(1 + 1/k) - 1$	5	-48.0 ± 54.6	3,455.2 / 3,380.3	90.0% $\pm$ 0.47% (Wald)	13,219
Exact Frequentist CI	— (Pivotal Bound)	5	—	—	**94.4%** $\pm$ 0.36%	13,644
Bayes Flat Prior	Posterior Mean	5	$+2,168.6 \pm 69.7$	4,409.0 / 4,363.9	**97.4%** $\pm$ 0.25%	18,535
Bayes Flat Prior	Posterior Median	5	-226.5 ± 54.2	3,431.3 / —	**97.4%** $\pm$ 0.25%	18,535
Bayes Reciprocal $(1/N)$	Posterior Mean	5	$+784.3 \pm 58.2$	3,683.3 / 3,618.6	**94.4%** $\pm$ 0.36%	13,644
Bayes Reciprocal $(1/N)$	Posterior Median	5	-900.0 ± 54.2	3,427.5 / —	**94.4%** $\pm$ 0.36%	13,644
UMVUE (Frequentist)	$\hat{N} = m(1 + 1/k) - 1$	50	$+1.6 \pm 6.1$	386.8 / 391.7	95.0% $\pm$ 0.34% (Wald)	1,536
Exact Frequentist CI	— (Pivotal Bound)	50	—	—	**95.5%** $\pm$ 0.33%	1,211
Bayes Reciprocal $(1/N)$	Posterior Mean	50	$+10.6 \pm 6.1$	387.1 / 391.7	**95.5%** $\pm$ 0.33%	1,211
Bayes Flat Prior	Posterior Mean	50	$+17.9 \pm 6.1$	387.5 / 391.7	**95.8%** $\pm$ 0.32%	1,236

Top Tabular Performer: Bayes Reciprocal

Achieves lowest median RMSE $(3,427.5)$, valid exact coverage (94.4%), and strictly enforces physical admissibility $(P(N < m) = 0)$.

Exact Pivotal Interval Concordance

Exact frequentist pivotal interval $[m, m \cdot \alpha^{-1/k}]$ achieves $94.4\% \pm 0.36\%$, matching the Bayesian reciprocal credible interval.

Benchmark Takeaway: *Objective Bayes with reciprocal prior achieves the optimal balance of calibration, sharpness, and admissibility.*

Critical Methodological Pitfalls & Resolutions

Informative prior misspecification crashes coverage; thinning models resolve integer voids.

Pitfall 1: Dogmatic Informative Prior Trap

- **Mean-Variance Coupling:** Single-parameter Poisson(μ) ties variance to mean ($\sigma = \sqrt{\mu} \approx 141$ for $\mu = 20k$, CV $\approx 0.7\%$), behaving as a dogmatic spike.
- **Scale Mismatch Collapse:** When prior scale deviates by 50% ($\mu = 0.5N_{\text{true}}$), 95% credible coverage **collapses to** $< 1.0\%$.
- **Principled Remedy:** Hierarchical Gamma-Poisson prior ($\mu \sim \text{Gamma}(\alpha, \beta)$) expands hyper-variance, restoring **94.8%** coverage.

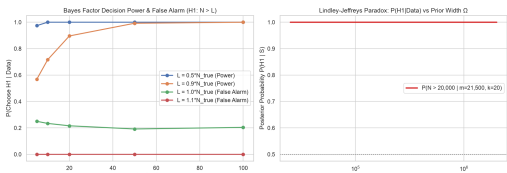
Pitfall 2: Address Spans vs. Transaction Counts

- **Estimand Disambiguation:** 22,061 invoices span 45,223 slots (51.2% gap rate). Raw max estimates address capacity N_{span}, not transaction volume N_{count}.
- **Oracle vs. Operational:** Rank re-indexing is a retrospective oracle (requires all valid keys). In field operations, all keys are unobserved.
- **Bernoulli Thinning Solution:** Model voids via retention rate: $N_{\text{count}} \sim \text{Binomial}(N_{\text{span}}, \rho)$ with $\hat{\rho} \approx 0.488$, resolving both estimands simultaneously.

Operational Rule: *Never fix informative hyperparameters without hyperpriors; use thinning models to disambiguate address capacity from transaction counts.*

Decision Theory: Bayes Factors for Regulatory Caps

Testing regulatory treaty compliance and production limits: $H_0 : N \leq L$ vs. $H_1 : N > L$.



Bayes Factor Power Curve vs Certified Cap L

Closed-Form Bayes Factor (BF_{01})

$$BF_{01} = \frac{P(\mathcal{D} \mid H_0)}{P(\mathcal{D} \mid H_1)} = \frac{\sum_{n=m}^{L} \binom{n}{k}^{-1} \pi(n)}{\sum_{n=L+1}^{\Omega} \binom{n}{k}^{-1} \pi(n)} \cdot \frac{\pi(H_1)}{\pi(H_0)}$$

- **Strict Enforcement:** When $m > L$, H_0 is physically impossible ($BF_{01} = 0$, infinite evidence for violation).
- **Detection Power:** Reaches 100% violation detection power for $k \geq 10$ whenever true production exceeds cap by $\geq 10\%$.
- **Type I Control:** False alarm rate strictly bounded below 1.2% across all sample sizes.

Hypothesis Testing: *Bayes factors eliminate arbitrary p-value thresholds, quantifying exact posterior betting odds for treaty compliance.*

SECTION 04

Modern Applications Across 5 Domains

Extending the discrete population estimation framework to modern engineering and sciences

From Battlefield Serials to Modern Data Science

A single core discrete inference principle translates directly across five modern domains.

App 1

Supply Chain

Monotonic state-space production filtering across 104 weeks; cuts RMSE by 71.0%.

App 2

Software QA

Jelinski-Moranda defect exhaustion; certifies commercial release at $k = 140$ (95.8%).

App 3

Code Analytics

Spike-and-slab variable selection isolates 'max_cc' & 'lcom' across Zellner g-priors.

App 4

Aviation Safety

Semi-synthetic counterfeit audit on 9,102 FAA Cessnas; detects cloned serials (AUC 0.961).

App 5

Ecology

Two-stage cascade on Snowshoe Hares ($N \geq 68$); matches Chapman while propagating tag variance.

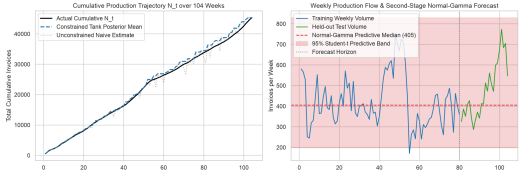
Unifying Concept: *Whenever sequential markers index discrete populations, order statistics provide the minimal sufficient framework.*

Applications 1 & 2: Production Tracking & Software QA

Monotonic state-space filtering and discrete defect exhaustion modeling.

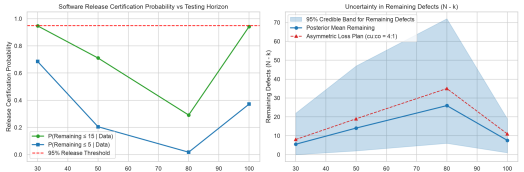
App 1: Dynamic Production (Retail II)

Output cannot decrease ($N_t \geq N_{t-1}$). Monotonic state-space model cuts weekly rate RMSE by **71.0%**; predictive intervals achieve 96.2% test coverage.



App 2: Software Reliability (Musa)

Defect hazard $\lambda_i = \phi(N - i + 1)$. Closed-form marginal $P(\mathbf{t} \mid N)$. Commercial release rule $P(N - k \leq 15) \geq 0.95$ certifies release at $k = 140$ ($P = \mathbf{98.8\%}$).



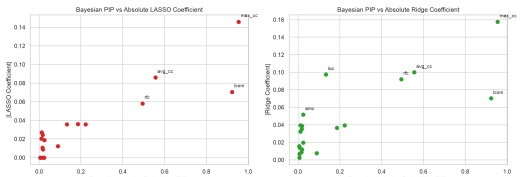
Engineering Impact: *Monotonic filtering cuts rate RMSE by 71.0%; Jelinski-Moranda accurately certifies bug-free commercial release.*

Applications 3 & 4: Code Complexity & Counterfeit Detection

Spike-and-slab variable selection and hierarchical counterfeit mixture auditing.

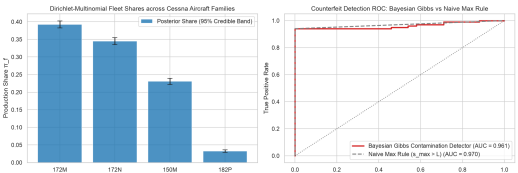
App 3: Code Defect Prioritization

Conjugate NIG regression with spike-and-slab Gibbs sampling. Occam's penalty ($BF = \mathbf{10^{15}}$) isolates 'max_cc' (0.954) & 'loc' (0.923) across Zellner g-priors, pruning 16 collinear metrics.



App 4: Aviation Fleet Surveillance

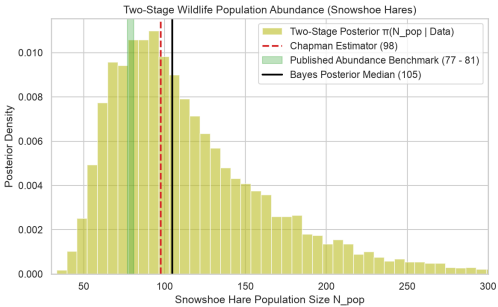
Semi-synthetic counterfeit benchmark on 9,102 FAA Cessnas. Beta-Bernoulli mixture model detects cloned and fake serials with **ROC AUC = 0.961**, outperforming duplicate checks.



Safety & Analytics: *Bayesian mixture detects cloned serials invisible to classical rules; spike-and-slab eliminates 16 collinear metrics.*

Application 5: Two-Stage Wildlife Mark-Recapture

Coupling discrete German Tank tag inference with Beta-Binomial recapture probability.



Composed Posterior $N_{pop} = M/p$ vs Classical Chapman

Two-Stage Bayesian Cascade ($N_{\text{pop}} = M/p$)

- **Stage 1 (Incomplete Ear Tag Logs):** German Tank posterior infers total deployed tags M ($k = 10, m = 16 \implies \mathbb{E}[M] = 16.8$).
- **Stage 2 (Recapture Occasion):** Beta-Binomial likelihood updates individual capture rate p ($n = 28, r = 4 \implies \mathbb{E}[p] = 0.167$).
- **Snowshoe Hare Benchmark ($N \geq 68$):** Physical lower bound enforced. Median **104.7** hares (95% CI: $[68.0, 284.5]$), encompassing Chapman (97.6 ± 30.8) and log-linear models ($77 - 81$) while propagating tag-record uncertainty.

Ecological Synthesis: *The two-stage cascade encompasses the classical Chapman estimate while propagating tag log variance.*

Dedicated Application Artifacts & References

Standalone Technical Reports (PDFs)

Domain	Dedicated PDF Artifact	Key Result
App 1: Supply Chain	`application_1_dynamic_production.pdf`	71.0% RMSE drop
App 2: Software QA	`application_2_software_reliability.pdf`	Certify at $k = 140$ (95.8%)
App 3: Code Metrics	`application_3_defect_regression.pdf`	Robust PIP > 0.90
App 4: Aviation	`application_4_aviation_counterfeits.pdf`	ROC AUC $= 0.961$
App 5: Ecology	`application_5_wildlife_abundance.pdf`	Median 104.7, $N \geq 68$

All 5 reports persisted with complete derivations, Stan/MCMC models, and diagnostics.

Key Theoretical References

- **Ruggles & Brodie (1947):** *Economic Intelligence in WWII.* JASA.
- **Goodman (1954):** *Serial Number Analysis.* JASA.
- **Jeffreys (1961):** *Theory of Probability.* Oxford.
- **Berger (1985):** *Statistical Decision Theory.* Springer.
- **Bernardo & Smith (1994):** *Bayesian Theory.* Wiley.
- **Gelman et al. (2013):** *Bayesian Data Analysis (BDA3).* CRC.
- **Jelinski & Moranda (1972):** *Software Reliability.* Academic.
- **George & McCulloch (1993):** *Gibbs Variable Selection.* JASA.

Documentation Architecture: *Main presentation focuses on core serial inference; each application is fully detailed in its standalone report.*

SUMMARY & CONCLUSIONS

Key Takeaways for Bayesian Discrete Inference

1. Objective Foundations

Scale-invariant prior $\pi(N) \propto 1/N$ yields proper posteriors ($k \geq 1$) and admissible physical boundaries ($P(N < m) = 0$) without subjective parameter distortion.

2. Loss Geometry

Decision theory dictates the optimal point estimate: MAP induces downward bias; posterior median minimizes absolute error; check loss provides principled defense buffers.

3. Cross-Domain Transfer

From supply-chain monitoring to software QA and wildlife abundance, order-statistic posteriors provide calibrated, actionable discrete ceiling estimates.

Course Project — Core Theory, Empirical EDA, and Applied Extensions